AI LAB ASSIGNMENT-8.1

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BATCH:44

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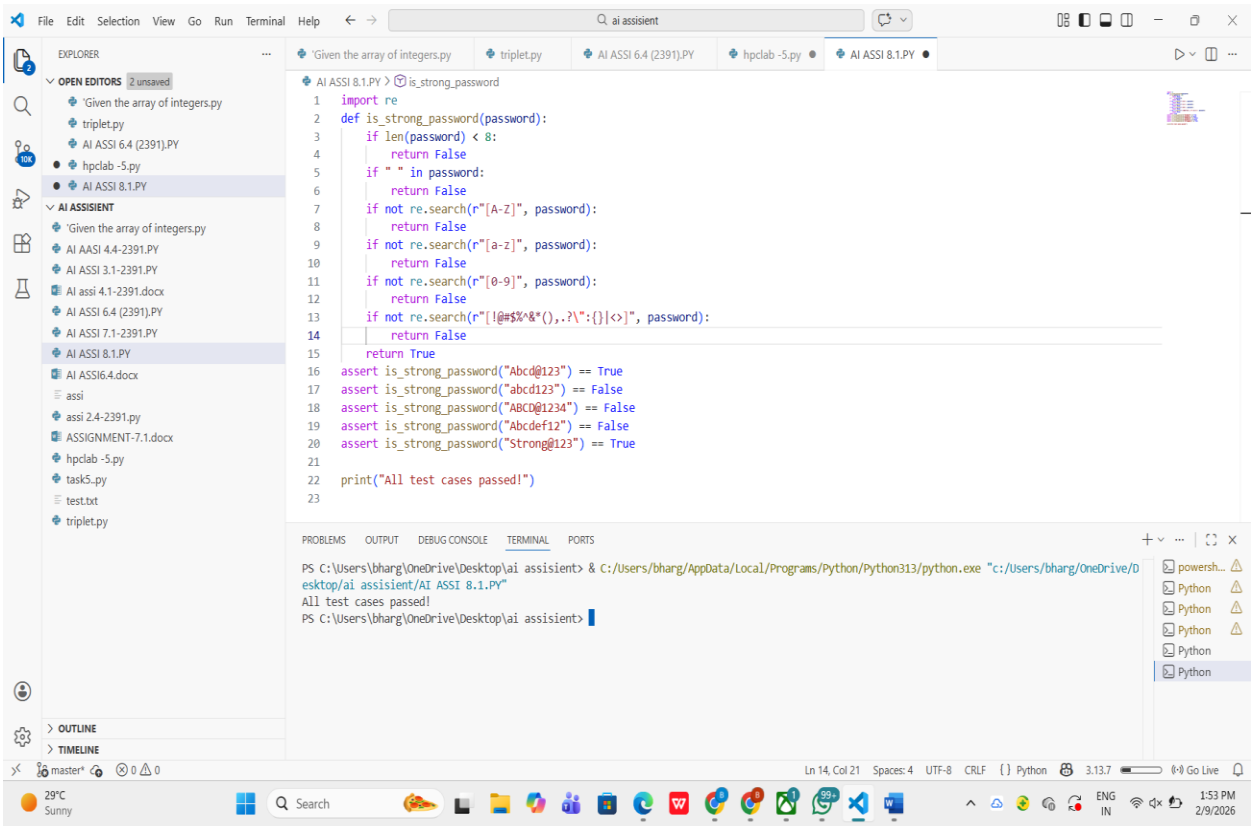
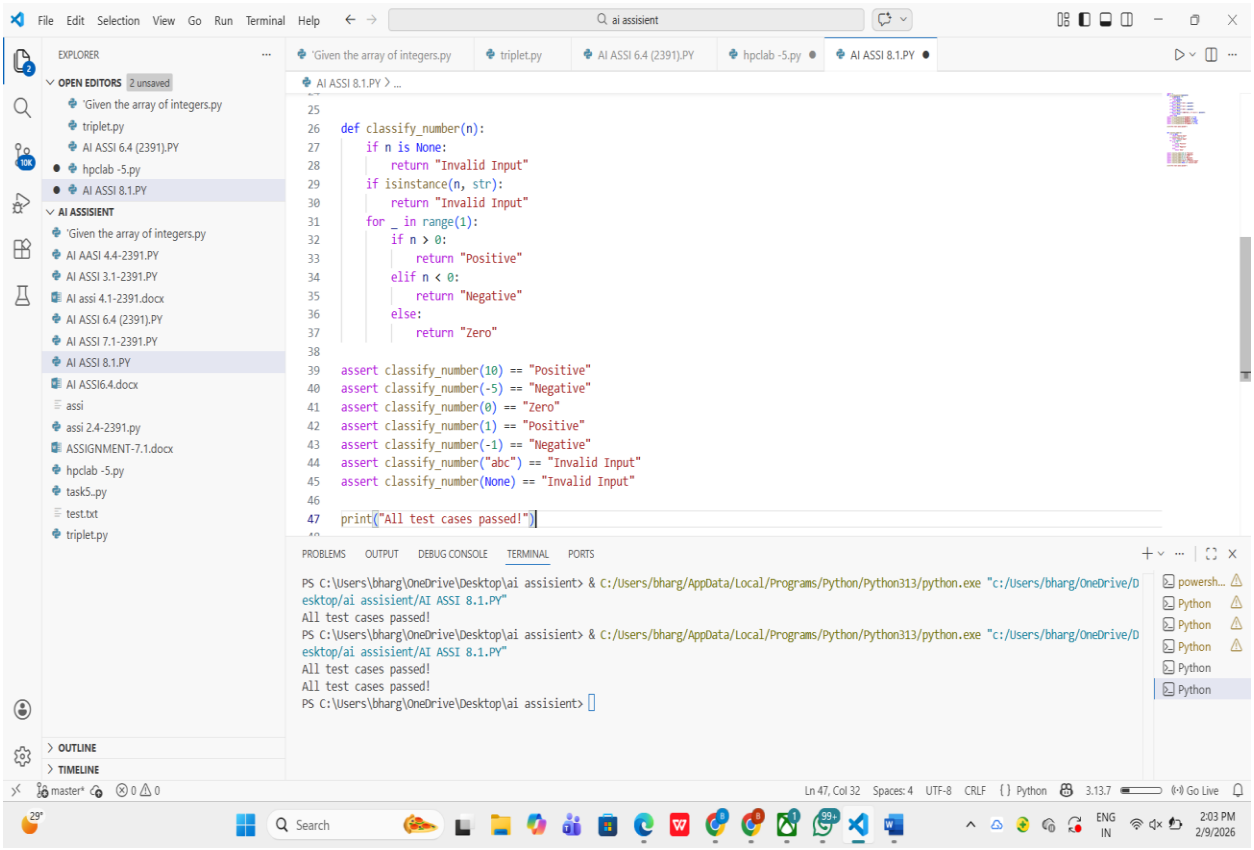
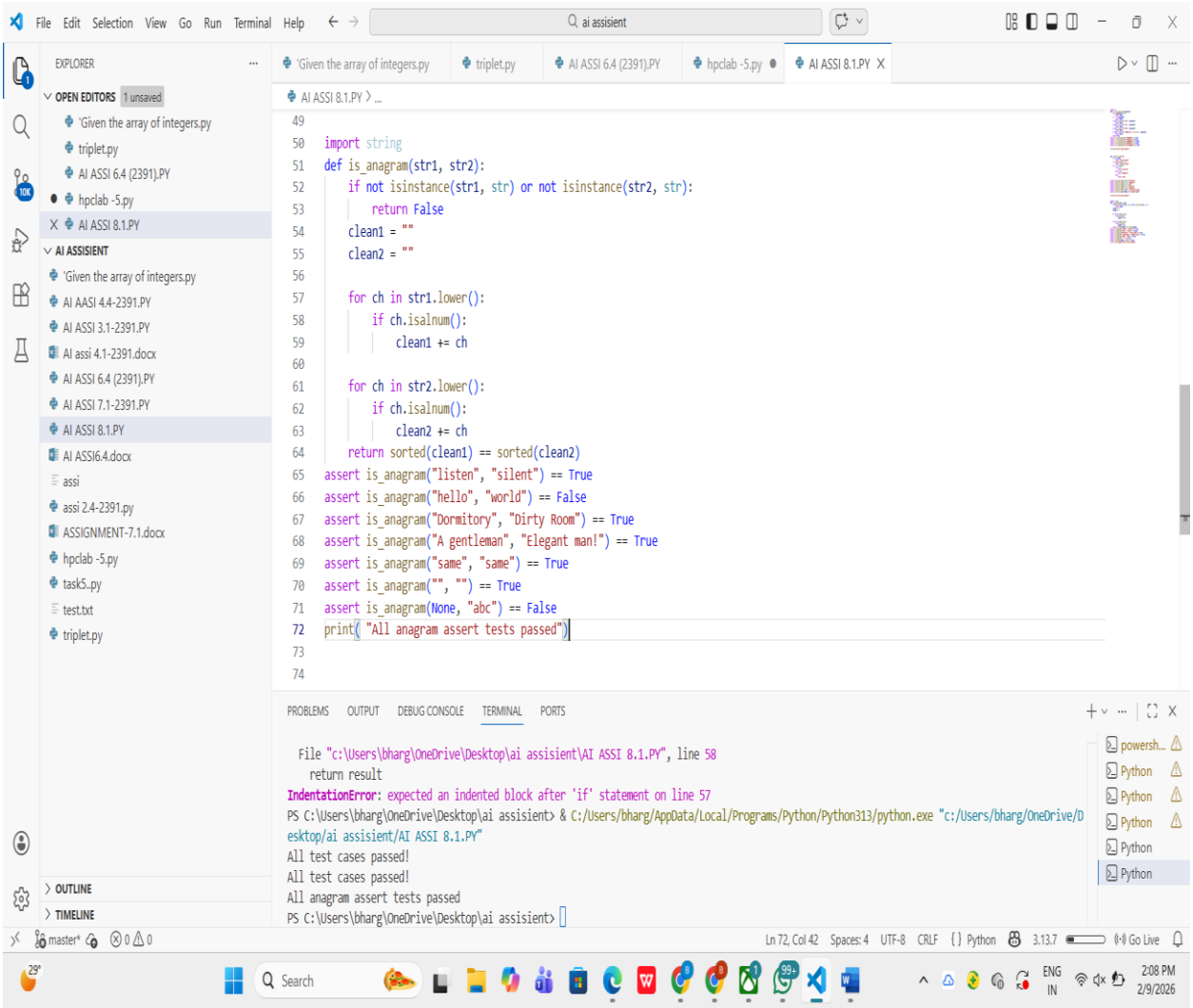
Task Description #1 (Password Strength Validator – Apply AI inSecurity Context)



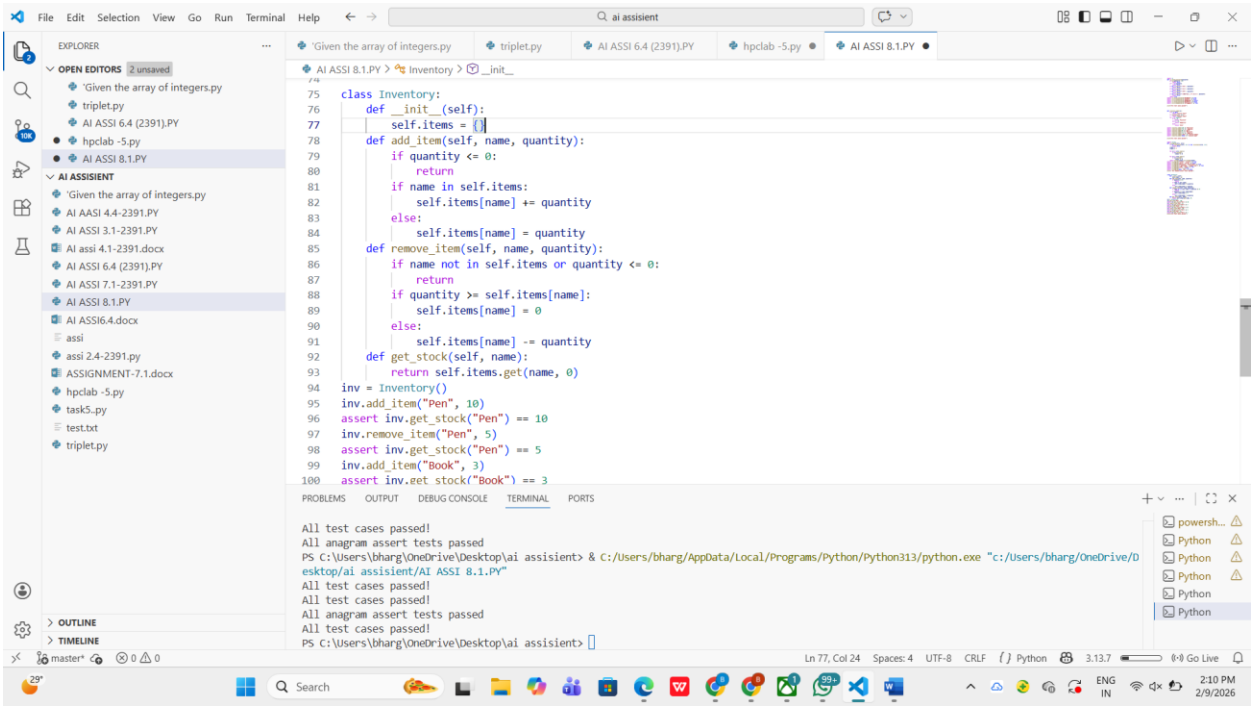
Task Description #2 (Number Classification with Loops – ApplyAI for Edge Case Handling)

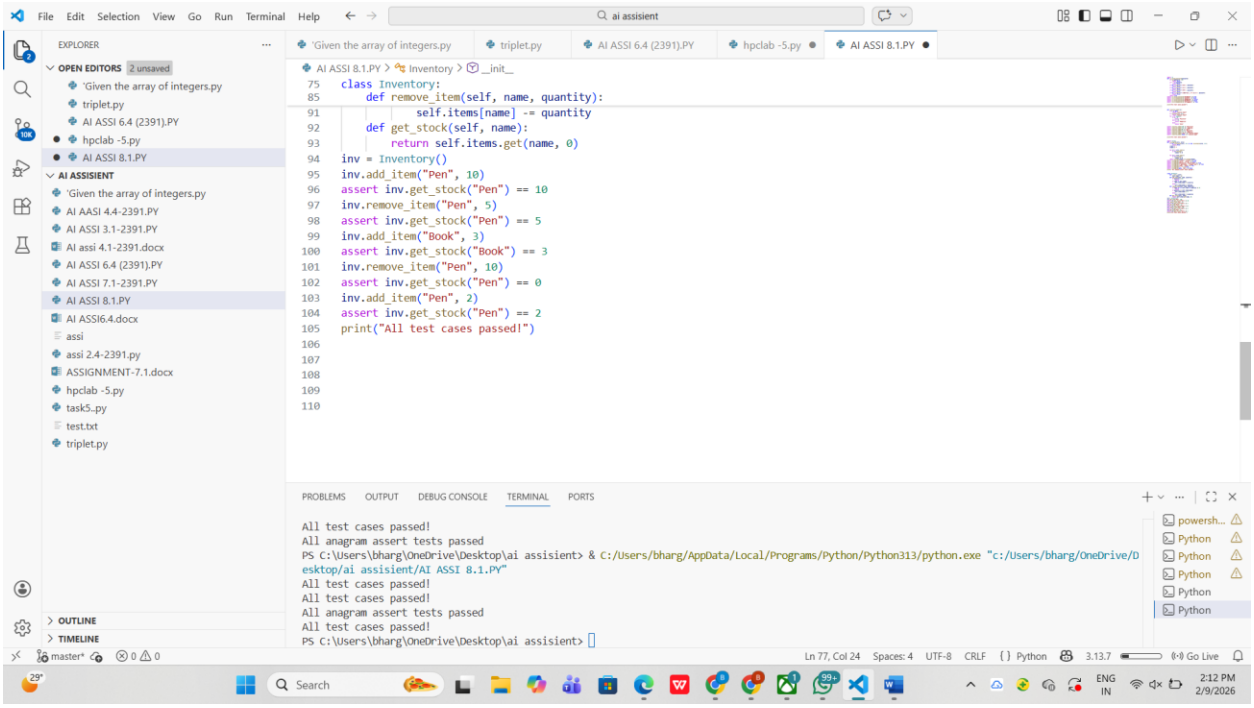


Task Description #3 (Anagram Checker – Apply AI for StringAnalysis)

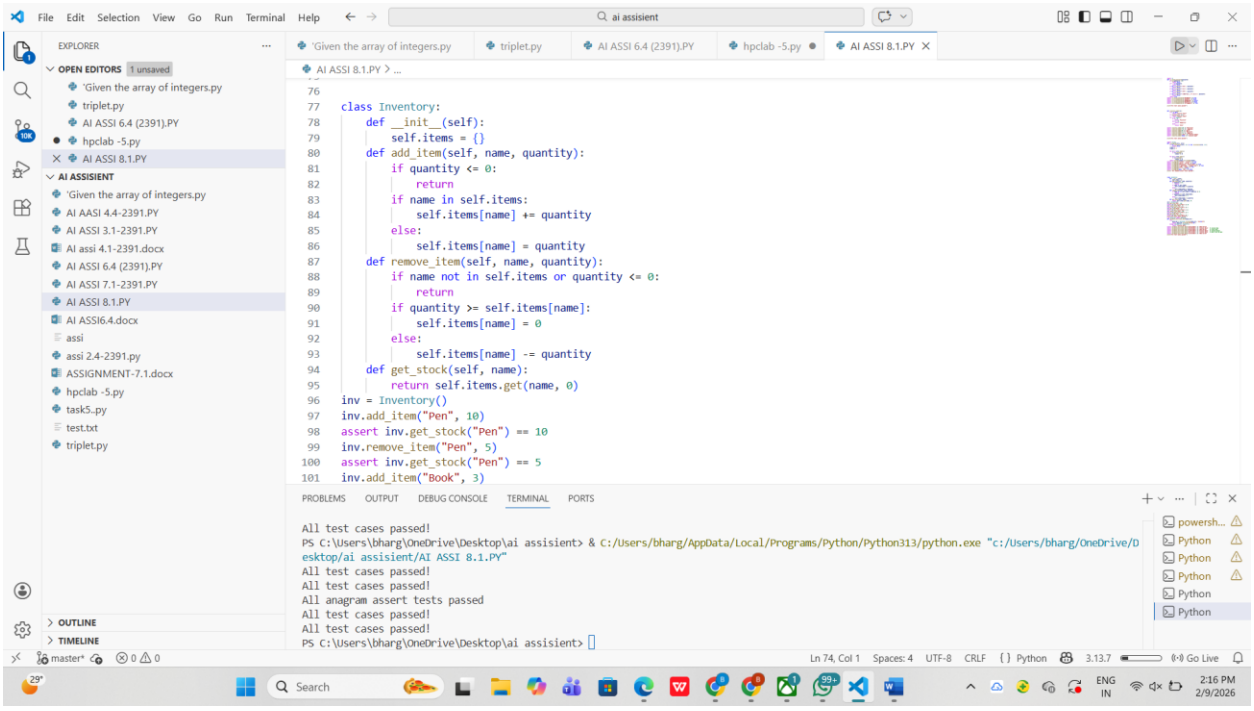


Task Description #4 (Inventory Class – Apply AI to Simulate Real-World Inventory System)





Task Description #5 (Date Validation & Formatting – Apply AI forData Validation)



```python
inv.add_item("Pen", 10)
assert inv.get_stock("Pen") == 10
inv.remove_item("Pen", 5)
assert inv.get_stock("Pen") == 5
inv.add_item("Book", 3)
assert inv.get_stock("Book") == 3
inv.remove_item("Pen", 10)
assert inv.get_stock("Pen") == 0
inv.add_item("Pen", 2)
assert inv.get_stock("Pen") == 2
print("All test cases passed!")
from datetime import datetime
def validate_and_format_date(date_str):
    try:
        date_obj = datetime.strptime(date_str, "%m/%d/%Y")
        return date_obj.strftime("%Y-%m-%d")
    except (ValueError, TypeError):
        return "Invalid Date"
assert validate_and_format_date("10/15/2023") == "2023-10-15"
assert validate_and_format_date("02/30/2023") == "Invalid Date"   # invalid day
assert validate_and_format_date("01/01/2024") == "2024-01-01"     # valid date
assert validate_and_format_date("13/01/2023") == "Invalid Date"   # invalid month
assert validate_and_format_date("04/31/2023") == "Invalid Date"   # April has 30 days
assert validate_and_format_date("12/25/2022") == "2022-12-25"
print("All test cases passed!")
```

```
All test cases passed!
PS C:\Users\bharg\OneDrive\Desktop\ai assisient> & C:/Users/bharg/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/bharg/OneDrive/Desktop/ai assisient/AI ASSI 8.1.PY"
All test cases passed!
All test cases passed!
All anagram assert tests passed
All test cases passed!
All test cases passed!
PS C:\Users\bharg\OneDrive\Desktop\ai assisient>
```